The `hawkdraw` package

A LaTeX package to generate vector graphic output based on l3draw and related LaTeX3 modules using a simple syntax

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1 Introduction

My sharp eyes spotted you, ground creature, already when you were still looking for this manual. Let me introduce myself: I am Victor, the drawing hawk. I epseically like drawing with LaTeX and since I value precision, using `l3draw` in combination with `l3fp` is my favourite way to draw. I created the `hawkdraw` package to make it easier for you to use the powerful drawing capabilities of the LaTeX3 kernel.

The `hawkdraw` package allows the user to use a simple syntax similar to Ti*k*Z to generate vector graphic output based on `l3draw` and related LaTeX3 modules such as `l3fp` which provides very precise calculations. The syntax is meant to be simple and flexible but also coherent and logical with the aim of allowing the user to quickly draw graphics. The package name relates to the quickness and agility of hawks and the preciseness of their eyesight.

The package intends to define a set of user-level commands that can be used for drawing simple graphics. It does not aim to be a substitute for the Ti*k*Z package or to offer a similar scope of application.

The package is currently still in an experimental stage. The code may be subject to fundamental and frequent changes. The author is grateful for reporting any bugs via GitHub at https://github.com/jasperhabicht/hawkdraw/issues. A site for asking questions about the package and for suggestions for improvement is available at https://github.com/jasperhabicht/hawkdraw/discussions.

! Note that the underlying code of the `l3draw` package which is about to be included as module to the L3 kernel is experimental as well and hence might change possibly breaking the functionality of this package. Although a row of fall-backs exist to ensure backward compatibility, in order to ensure the functionality of the `hawkdraw` package, it is highly recommended to use it with an up-to-date TeX distribution, especially with the most recent version of the `l3draw` package.

* E-mail: mail@jasperhabicht.de. I am grateful to Joseph Wright, Max Chernoff, Clea F. Rees, Mikael Persson Sundqvist and Jonathan P. Spratte who helped me improving the code and the math behind it.

2 Contents

3 Loading the package

To install the package, copy the package file `hawkdraw.sty` together with the module files ending with `.code.tex` into the working directory or into the `texmf` directory. After the package has been installed, the `hawkdraw` package is loaded by calling `\usepackage{hawkdraw}` in the preamble of the document.

The package can be used with PDFLaTeX, LuaLaTeX or XeLaTeX. The package does not load any dependencies, but it needs a LaTeX kernel of 1 June 2022 or newer. It is recommended to use the package with an up-to-date TeX distribution.

4 General remarks

Similar to Ti*k*Z, the `hawkdraw` package makes use of key-value pairs to set options for commands. Some options are available for almost all commands, other have very specific uses. In the descriptions of the options, this manual states which data type the relevant option assumes. The following data types exist:

Data type	Explanation
`boolean`	A boolean only allows the values `true` or `false`. If not stated otherwise, setting the key without a value is equivalent with setting the key to `true`.
`string`	A string is a concatenation of letters or numbers of any length. Typically, strings do not contain symbols. For example, strings are used for names of points, frames of nodes, patterns or arrow tips.
`float`	A floating-point number is a number without or with unit that uses a colon as decimal separator.
`dimension`	A dimension or length is a typical TeX length consisting of a floating point number and a length unit.
`tuple`	A tuple consists of two floating-point numbers without or with unit that are separated by a comma. A tuple may be enclosed by parenthesis.
`point`	A point is essentially a tuple, but it can also be a reference to a named node or, using a special syntax, denote a point in polar coordinates (see below).
`clist`	A comma-separated list is a list of strings that are separated by commas. Such a list can cosnist of no or only one item as well. If the relevant list can only have a maximum number of items, this is stated in the description of the relevant key. Note that a comma-separated list is typically not surrounded by parenthesis.
`options`	A key-value list consists of comma separated entities of keys that may be followed by an equals sign and a value. A value can be another key-value list.
`color`	This means a color definition as specified by the `l3color` module. Note that colors defined by the `xcolor` package have no effect when using the `hawkdraw` package.
none	Some keys do not accept any data type as they are only executing some function.

Keys in the `hawkdraw` package are divided into sets. Depending on where the keys are used in the document, only specific sets can be made use of. If a command or environment accepts specific sets of keys, the relevant prefix should be omitted.

\hawkdrawset{‹options›}